

E0.3 · E0.4 · E0.5 / HBN THEPRESENT / R5 VAL

What the REVE warm start actually buys

Three encoders on one subject-scaling axis. The random-init control separates two effects that were confounded in every earlier comparison: the warm start sets the *level*, and capacity sustains the *slope*.

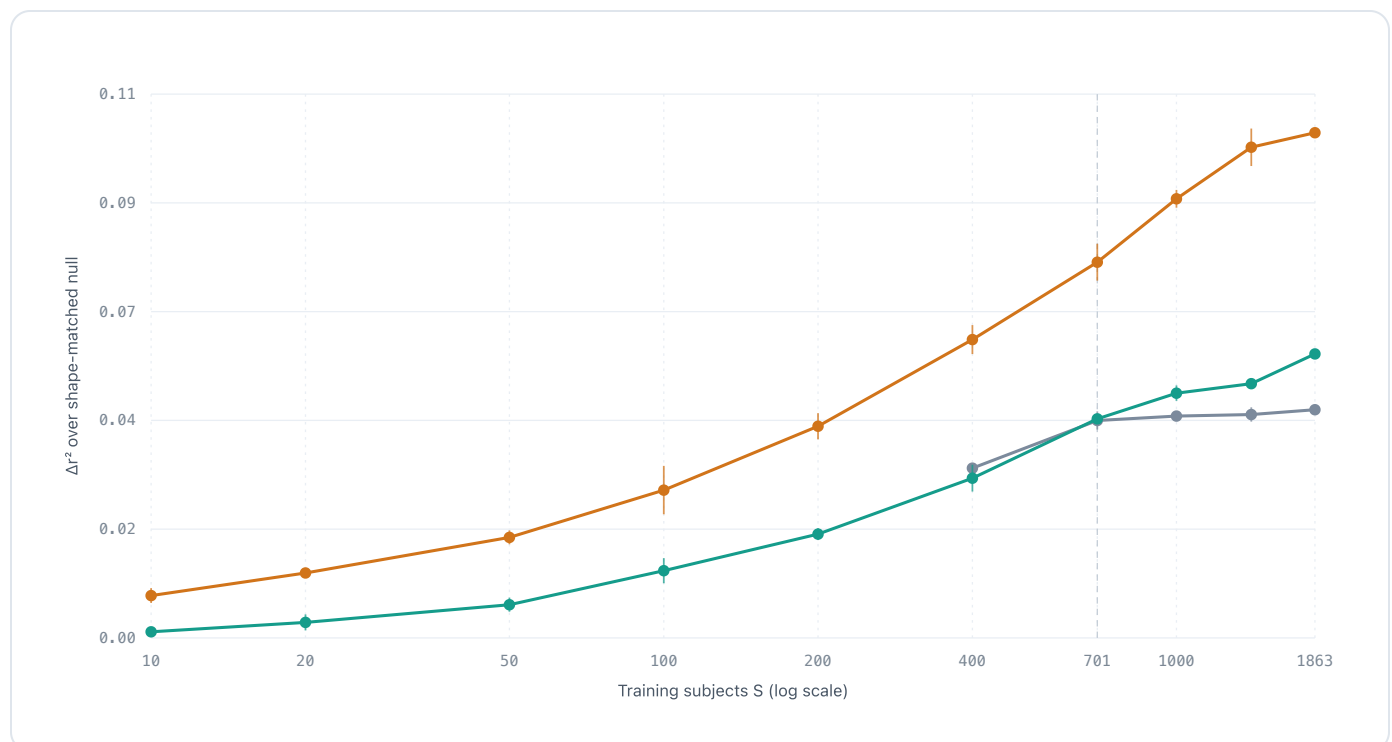
depth 12 · random init

depth 22 · random init

depth 22 · REVE warm start

Figure 1 — Overall stimulus decoding vs. training subjects

Mean Δr^2 across 12 stimulus features, each arm measured against a random encoder of its own shape. Error bars are ± 1 sd over three cohort draws; $S = 1863$ is the whole pool, so only one draw exists. Dashed line marks $S = 701$.



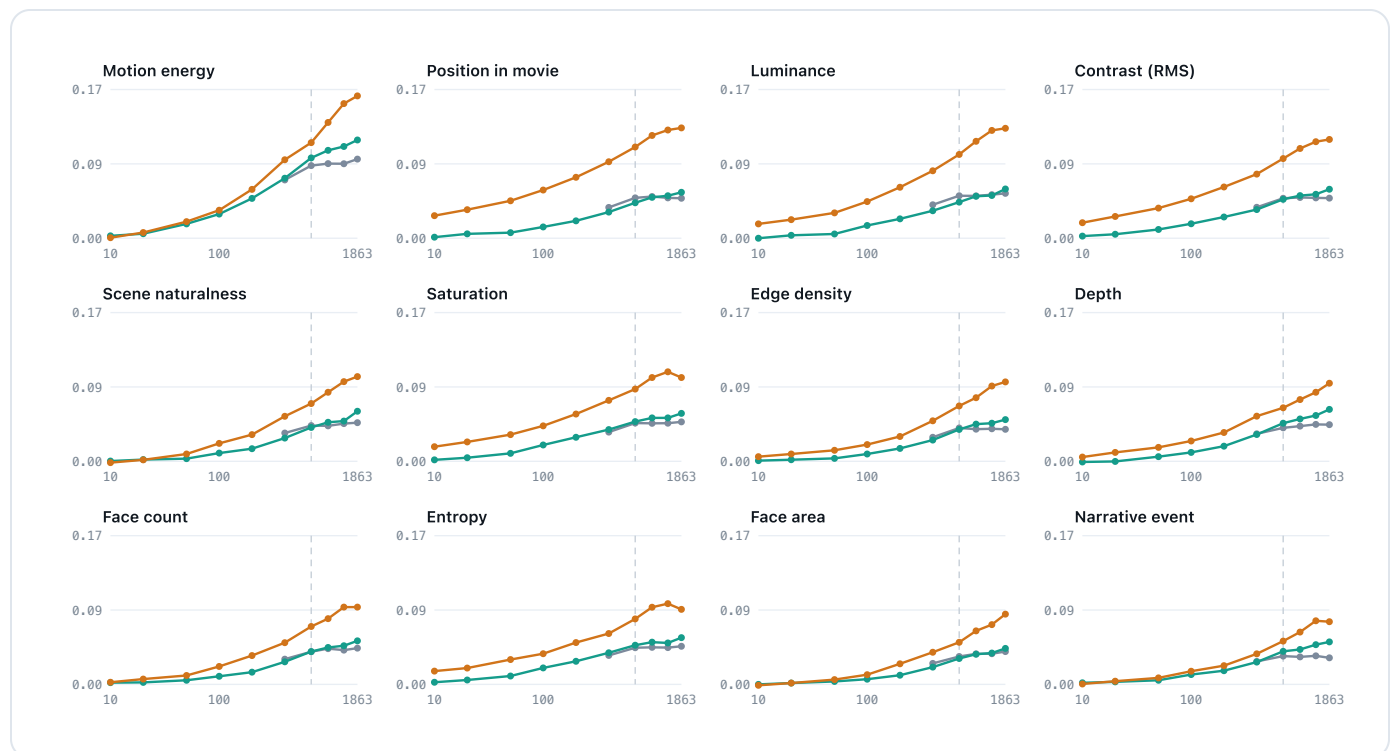
Depth 22 at random init tracks depth 12 up to ~700 subjects, then keeps climbing where depth 12 flattens. Both random arms sit far below the warm-started encoder at every scale.

The level is the warm start. Random-init depth 22 lands on the depth-12 curve — 0.03231 vs 0.03434 at $S = 400$, 0.04431 vs 0.04398 at $S = 701$ — despite 1.8× the parameters, 2× the tokens and 2× the gradient steps. The $\sim 1.7\times$ gap to E0.4 is the pretrained initialization, not capacity.

The slope is capacity. From 701→1863 depth 12 gains +4.9%, depth 22 at the same random init gains +29.6%, and the warm-started arm +34.4%. Local exponents over that range: **0.049** (d12), **0.265** (d22 random), **0.303** (d22 warm). The saturation near 700 subjects is a property of the depth-12 encoder, not of the subject axis.

Figure 2 — The same curve, one panel per feature

All 12 probe features share a common y-axis (0–0.17) so panels are directly comparable; ordered by warm-start signal at $S = 1863$. X ticks at 10 / 100 / 1863.



The ordering is stable across arms: motion energy carries the most decodable signal in all three, narrative event and face features the least. No feature reverses the arm ordering, and none saturates while the others climb — the warm start lifts the whole feature set rather than a subset.

Overall Δr^2 by scale

Mean across features, ± 1 sd over cohort draws.

S	D12 RANDOM	D22 RANDOM	D22 WARM
10	—	0.00123 ± 0.00026	0.00857 ± 0.00150
20	—	0.00313 ± 0.00162	0.01313 ± 0.00040
50	—	0.00672 ± 0.00144	0.02034 ± 0.00139
100	—	0.01360 ± 0.00255	0.02990 ± 0.00489
200	—	0.02100 ± 0.00121	0.04282 ± 0.00264
400	0.03434 ± 0.00057	0.03231 ± 0.00271	0.06034 ± 0.00293
701	0.04398 ± 0.00183	0.04431 ± 0.00082	0.07601 ± 0.00373
1000	0.04489 ± 0.00013	0.04951 ± 0.00155	0.08883 ± 0.00180
1400	0.04519 ± 0.00145	0.05142 ± 0.00066	0.09926 ± 0.00379
1863	0.04615	0.05743	0.10218

Per-feature Δr^2 at S = 1863

The whole training pool. Final column is the warm start's multiple over the matched random-init encoder.

FEATURE	D12 RANDOM	D22 RANDOM	D22 WARM	WARM / RANDOM
Motion energy	0.0904	0.1124	0.1626	1.45×
Position in movie	0.0456	0.0525	0.1261	2.40×
Luminance	0.0512	0.0563	0.1257	2.23×
Contrast (RMS)	0.0459	0.0560	0.1129	2.02×
Scene naturalness	0.0442	0.0572	0.0968	1.69×
Saturation	0.0451	0.0548	0.0957	1.75×

Edge density	0.0365	0.0476	0.0907	1.91×
Depth	0.0419	0.0593	0.0893	1.51×
Face count	0.0415	0.0499	0.0884	1.77×
Entropy	0.0435	0.0535	0.0858	1.61×
Face area	0.0374	0.0412	0.0804	1.95×
Narrative event	0.0304	0.0486	0.0717	1.47×

What these numbers do not say

- **E0.5 ran 8800 gradient steps to E0.4's 4400.** At 400 epochs 11 of 13 random-init cells with $S \geq 400$ were still improving at the final epoch, so the arms were compared at each one's own convergence rather than at equal compute. This makes the level result *stronger*: the random arm loses with double the budget.
- **E0.4's $S = 1863$ point is the seed-7 rescue** (0.10218), not the undertrained original (0.07549). With the original, that arm's 1400→1863 exponent reads -0.958 instead of $+0.102$.
- **$S = 1863$ is $n = 1$ in all three arms** — one draw of the whole pool exists, so that column carries no error bar.
- **This is one readout.** Every number here is the 12-feature CV probe on val. Retrieval on these same checkpoints has previously disagreed with this probe about saturation, and has not been run for E0.5. The saturation claim is about the probe, and should be written that way.
- **E0.4 and E0.5 differ in patch geometry as well as init** (200/20, forced by reve-base's weights) but E0.5 isolates init cleanly against E0.4, since those two share a shape exactly.

HBN *ThePresent*, train R1–R4 + R7–R10, val R5. Encoder shape embed 512, patch 200/20 (d22 arms) · patch 400/0 (d12). Nulls: $r^2 = 0.016572$ (depth 22), 0.014127 (depth 12). Probe: 5-fold CV by recording, 12 features.

